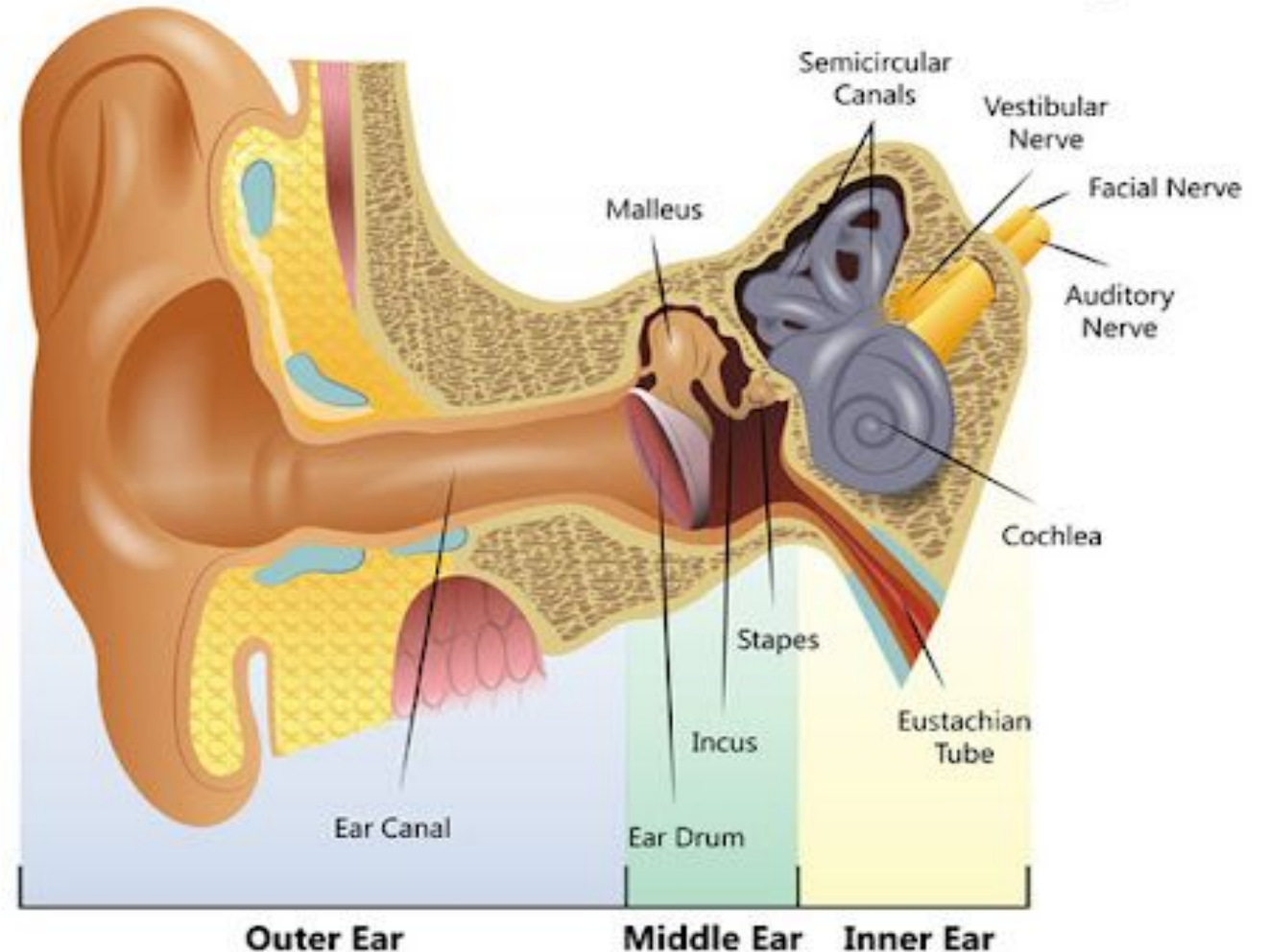


Gross Anatomy of the EAR



Introduction:

- Ear is an organ of hearing and is also concerned in maintaining the equilibrium of the body.
- It consists of three parts viz, the **external ear**, the **middle ear** and the **internal ear**.

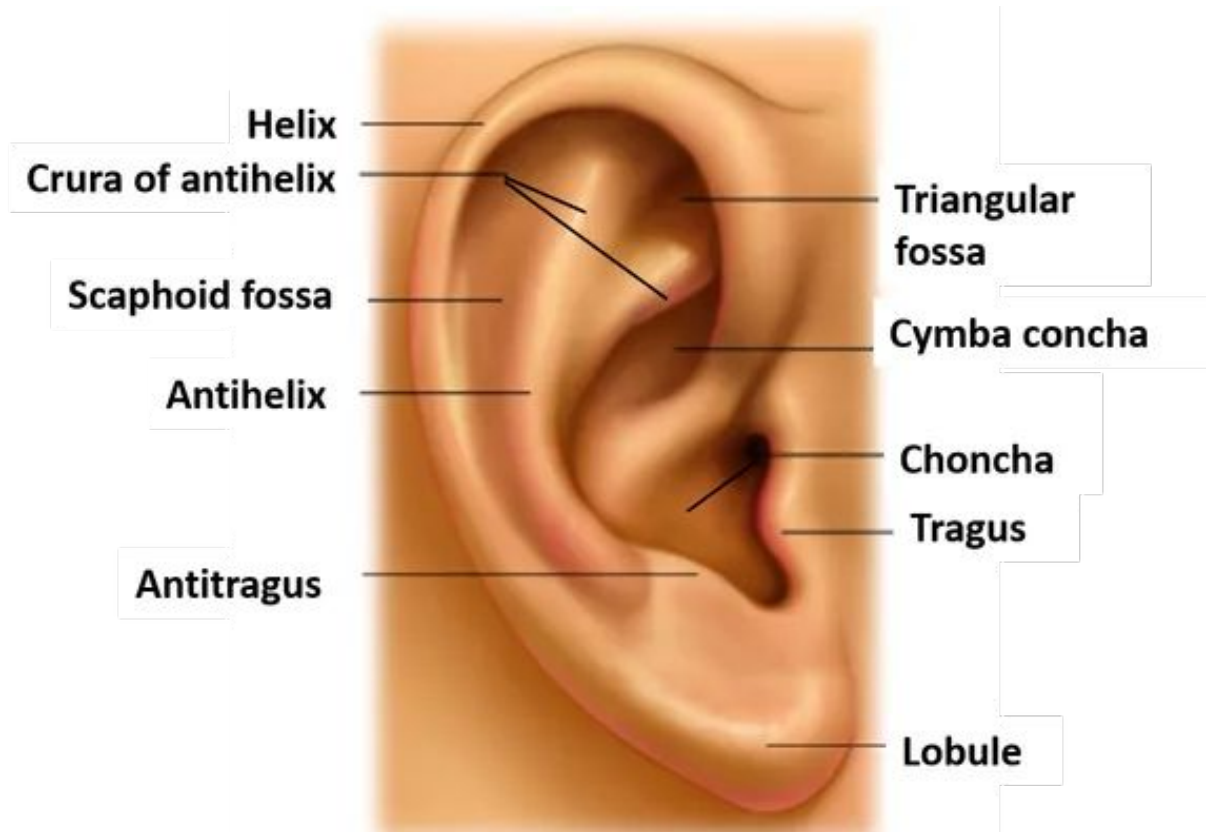


The External Ear:

The external ear consists of: **auricle/pinna** and **external acoustic meatus**

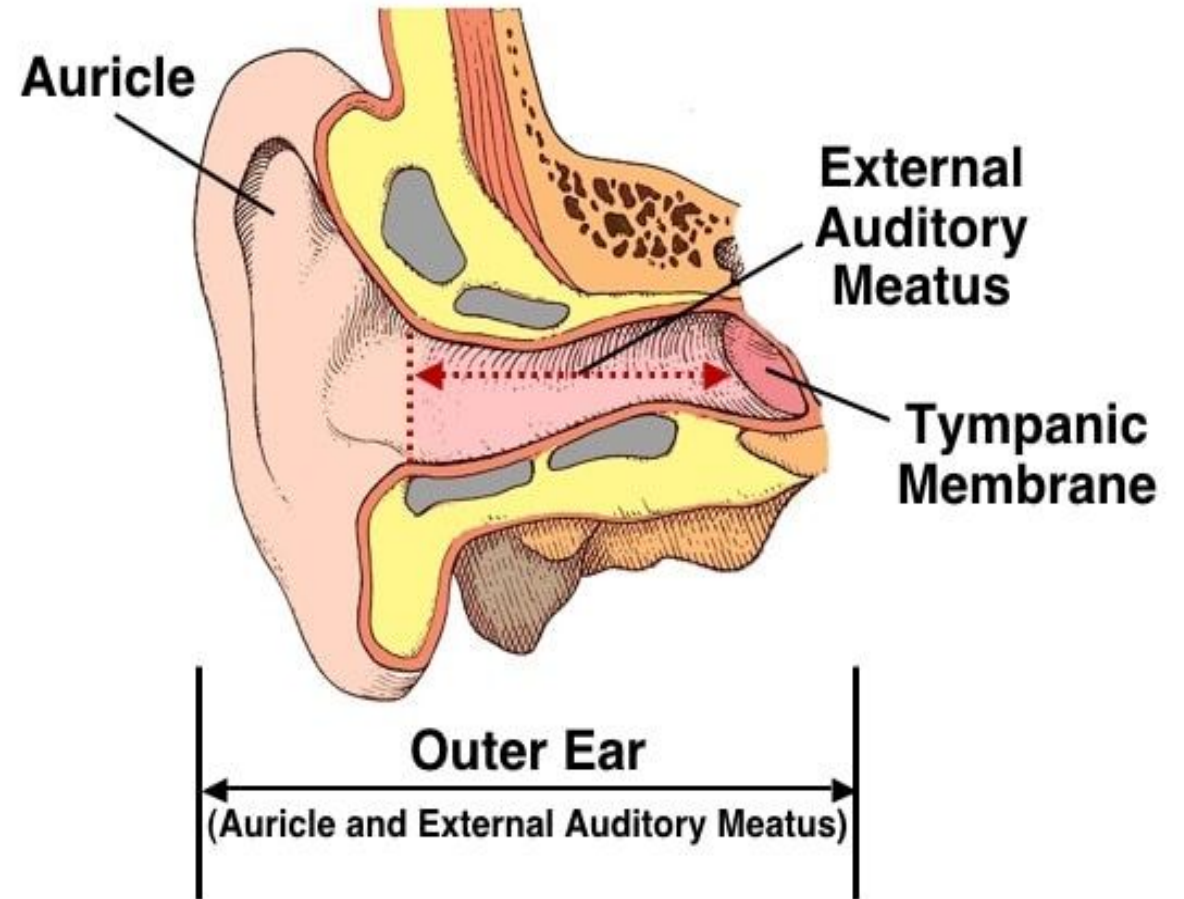
- **Auricle/pinna:** is made up of elastic cartilage covered by skin.
- Lower part of auricle is soft containing connective tissue covered by skin is called **lobule**.
- **Blood supply:** posterior auricular and superficial temporal arteries.
- **Nerve supply:** root of the auricle is supplied by auricular branch of vagus nerve and lower one-third is supplied by great auricular nerve whereas upper 2/3rd of lateral surface is supplied by auriculotemporal nerve and that of medial surface by lesser occipital nerve.

Parts of the Auricle:



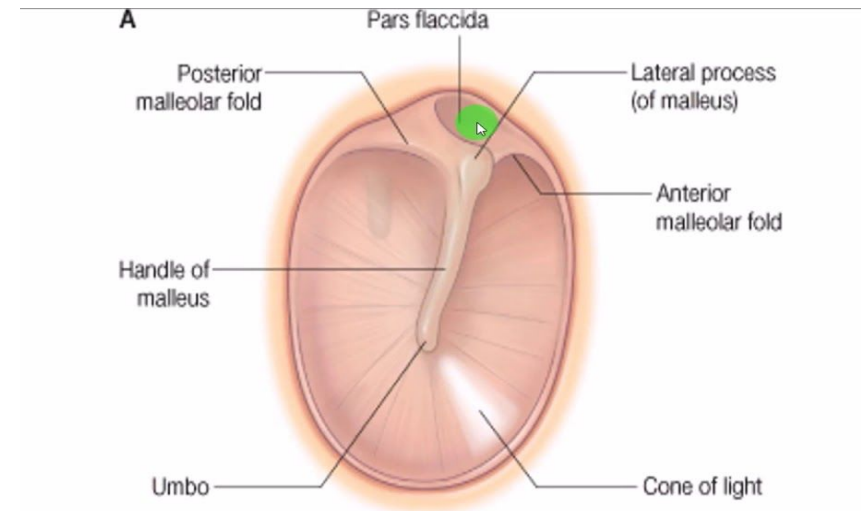
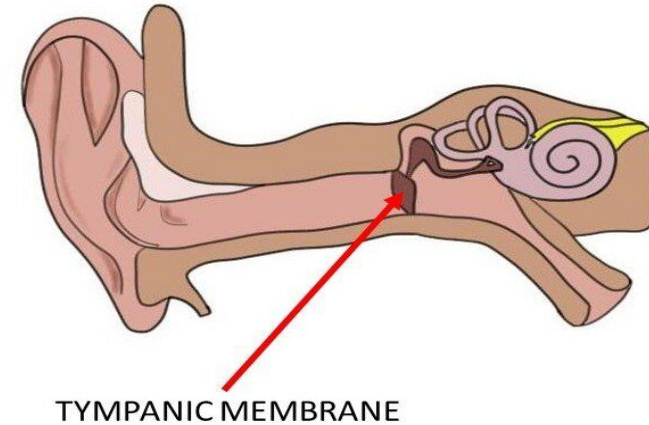
The External Ear:

- **External acoustic /auditory meatus:** transmits sound waves from concha to tympanic membrane.
- S-shaped. Its medial $\frac{2}{3}$ rd is bony and lateral $\frac{1}{3}$ rd is cartilaginous.
- Bony part is made up of tympanic plate of temporal bone and cartilaginous part is made up of elastic cartilage.
- Lining skin consists of hairs, sebaceous glands and ceruminous/wax glands.
- **Blood supply:** outer part is supplied by superficial temporal and inner part is supplied by maxillary artery.
- **Nerve supply:** anterior half is supplied by auriculotemporal and posterior half is supplied by auricular branch of vagus nerve.



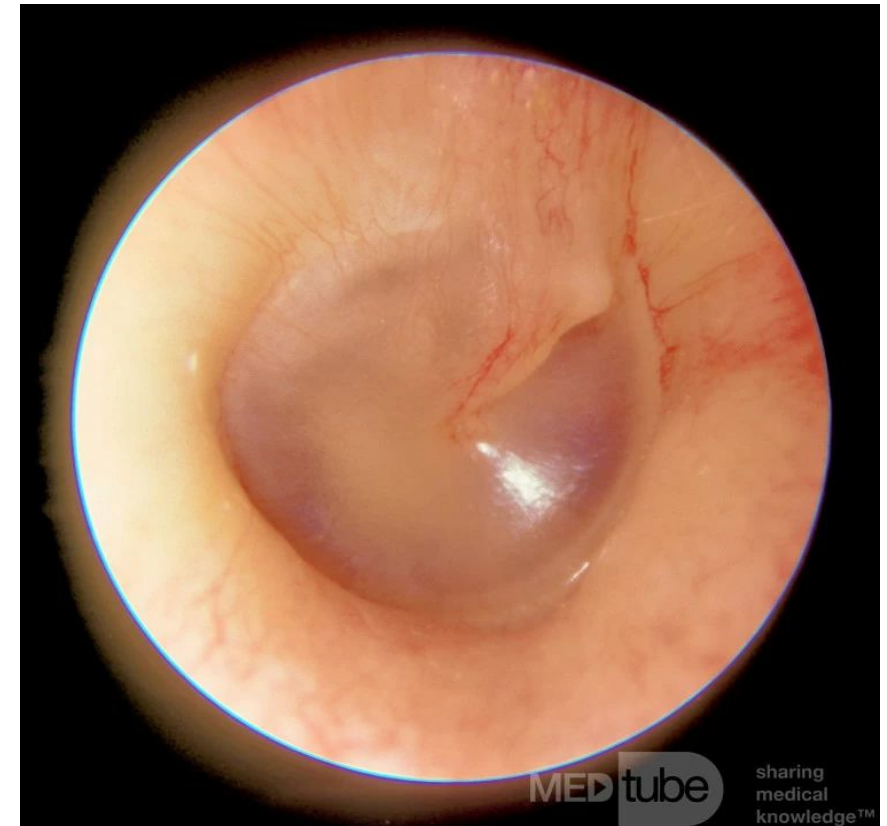
Tympanic membrane (ear-drum):

- It is a thin, translucent, pearly white partition between external acoustic meatus and middle ear.
- It is oval in shape, its outer surface is concave and covered by skin whereas inner surface is convex and gives attachment to handle of malleus.
- The point of maximum convexity is called umbo.
- Tympanic membrane is fixed at its circumference to the tympanic sulcus of temporal bone. The sulcus is deficient superiorly.
- The greater part of tympanic membrane is tightly stretched, called PARS TENSA.
- The part between anterior & posterior malleolar folds is loose, called PARS FLACCIDA. It is more prone to *perforations*.
- It is made up of three layers viz, outer cuticular layer of skin, middle fibrous layer and inner mucous layer lined by low ciliated columnar epithelium.



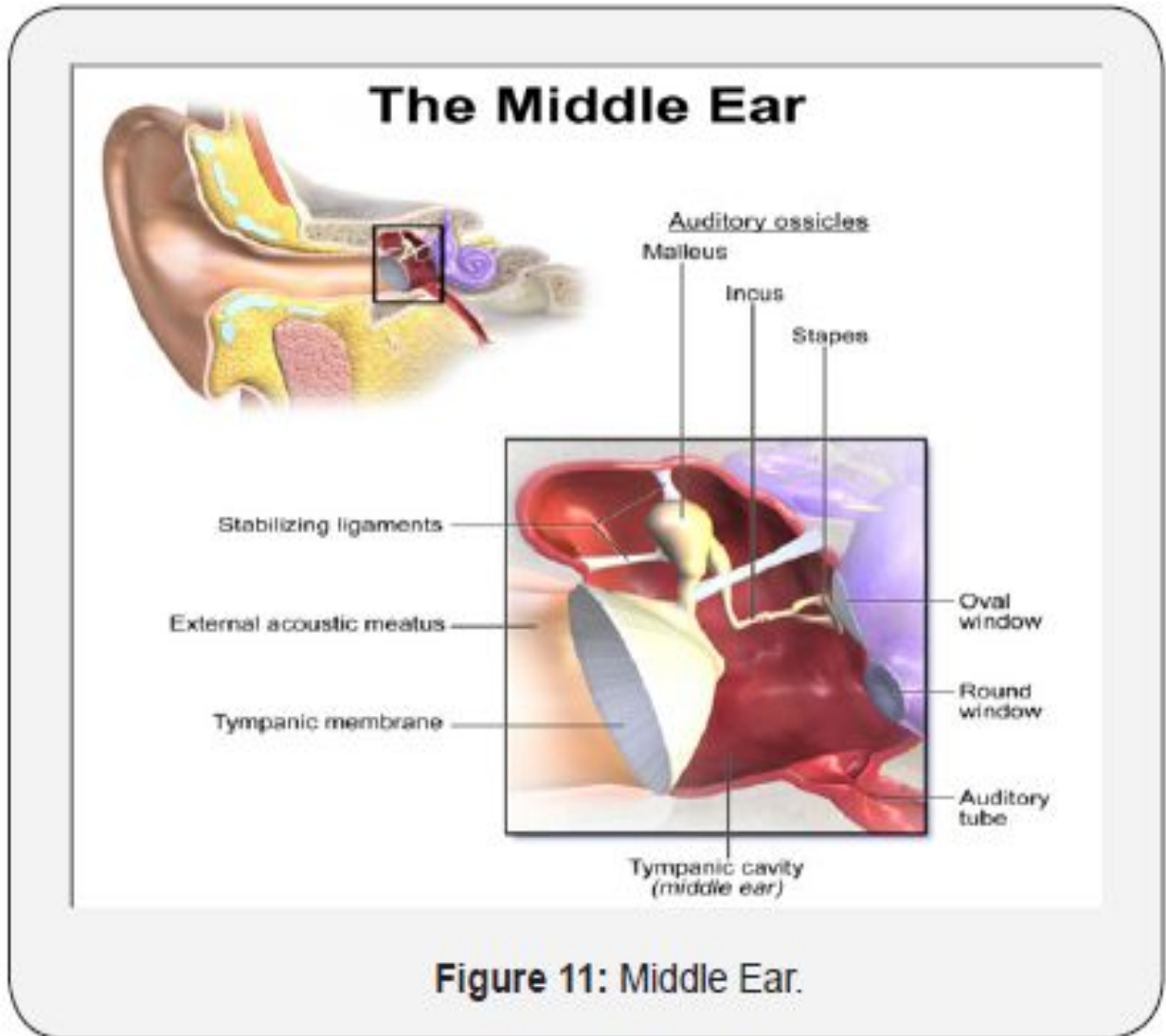
Tympanic membrane:

- **Blood supply:** outer surface is supplied by deep auricular branch of maxillary artery and inner surface is supplied by anterior tympanic branch of maxillary artery and posterior auricular artery.
- **Venous drainage:** outer surface drains into external jugular vein and inner surface into transverse sinus and venous plexus around auditory tube.
- **Nerve supply:** outer surface is supplied by auriculotemporal nerve and auricular branch of vagus nerve whereas inner surface by tympanic branch of glossopharyngeal nerve through tympanic plexus.



The Middle Ear (tympanic cavity):

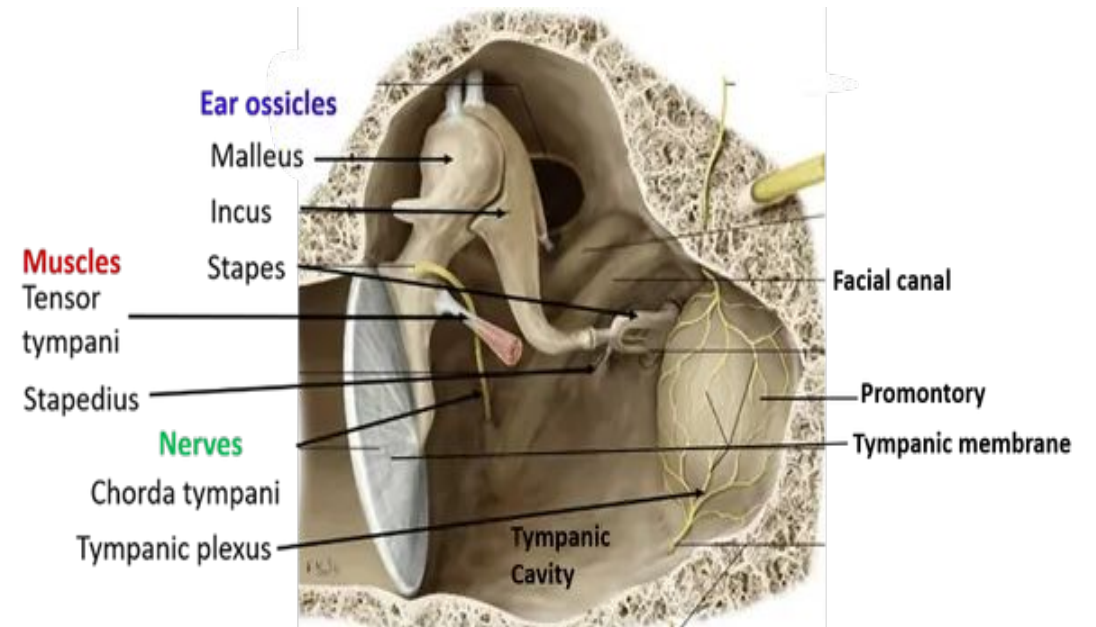
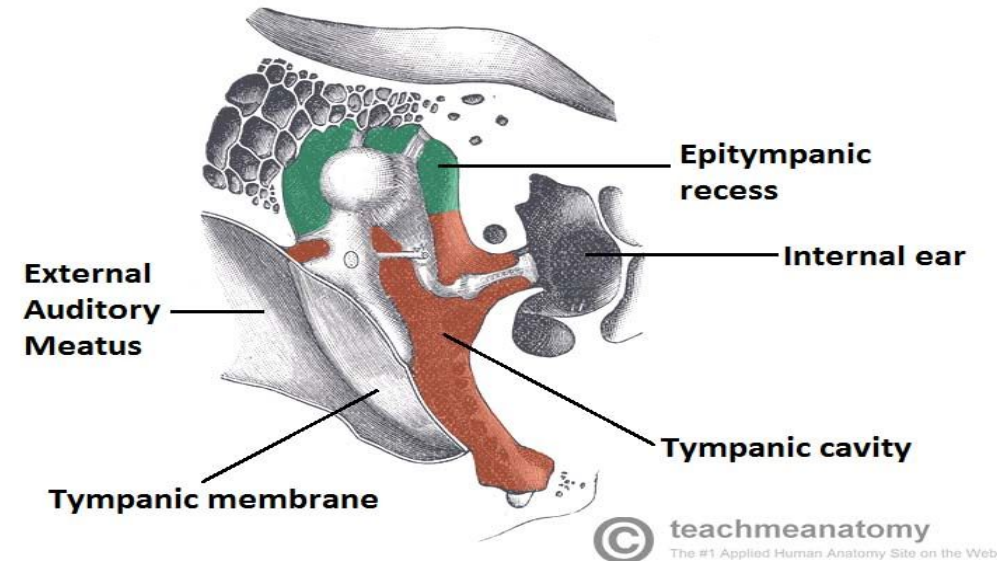
- It is a narrow air-filled space located in the petrous part of temporal bone between external ear and internal ear.
- It is cube-like in shape but biconcave when seen in coronal section.
- It communicates anteriorly with the nasopharynx via auditory tube and posteriorly with the mastoid antrum through aditus to the mastoid antrum.



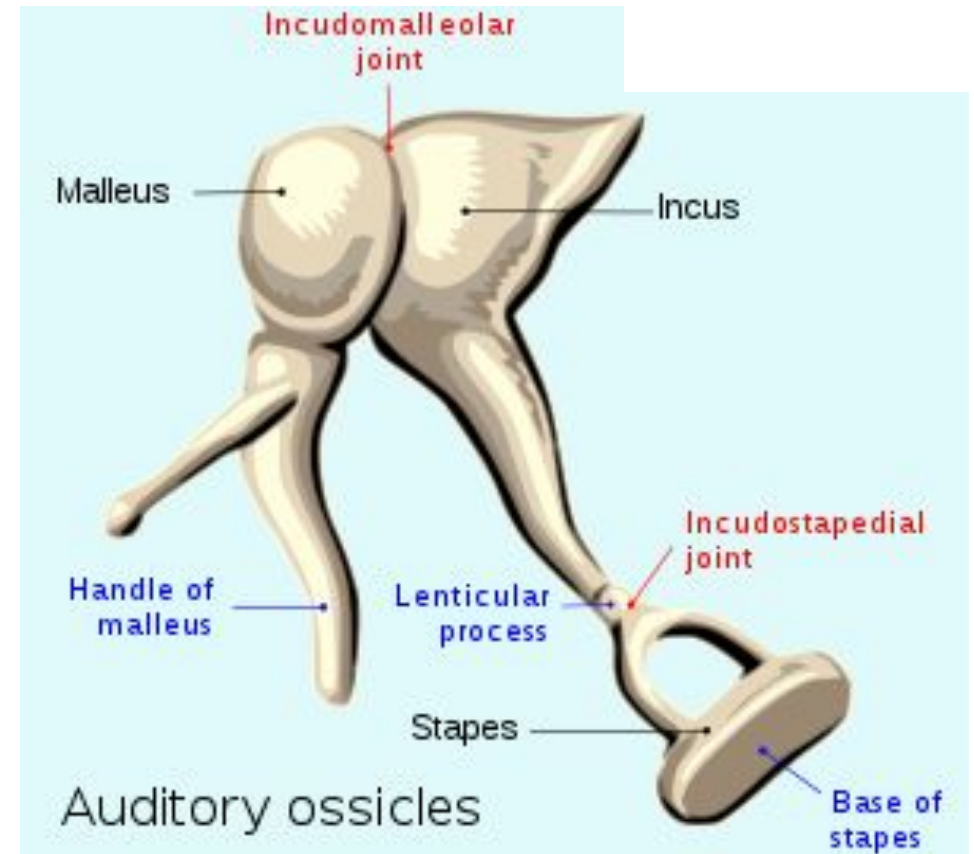
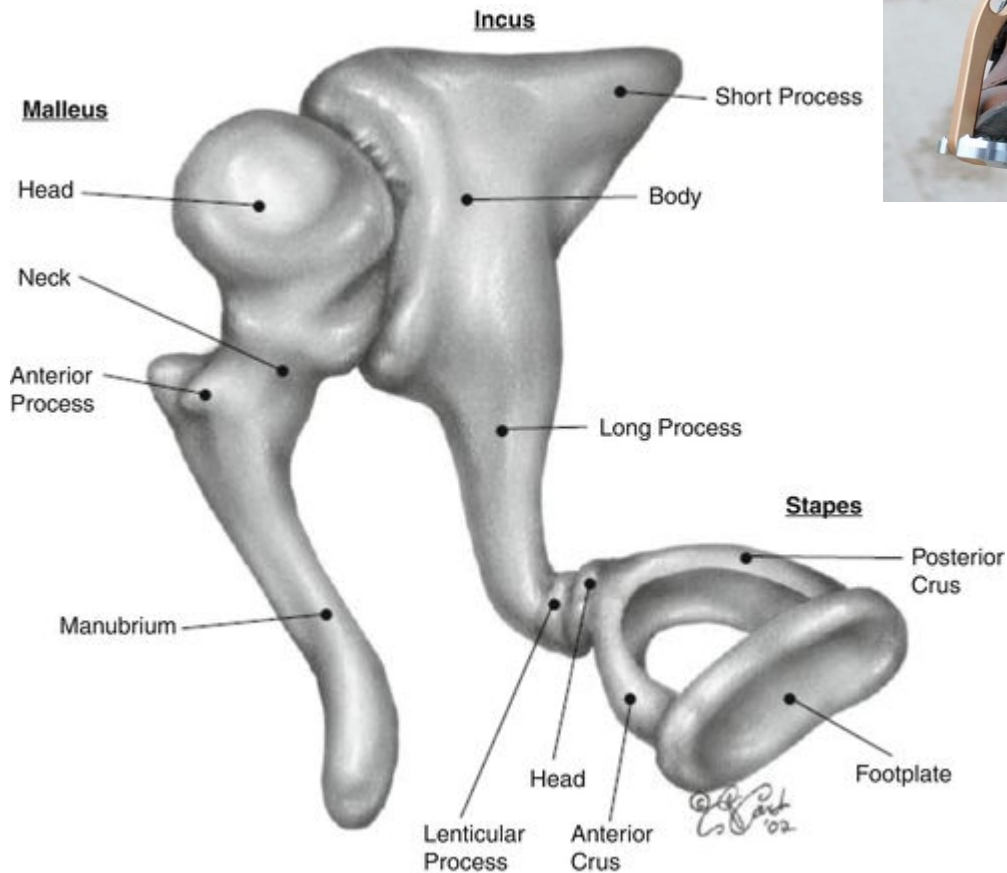
Contents of the middle ear cavity:

The middle ear cavity contains:

1. Three small bones/auditory ossicles namely, malleus, incus and stapes.
2. Ligaments of ear ossicles.
3. Two muscles, the tensor tympani and the stapedius.
4. Blood vessels.
5. Chorda tympani branch of facial nerve and tympanic plexus.



The Ear Ossicles:



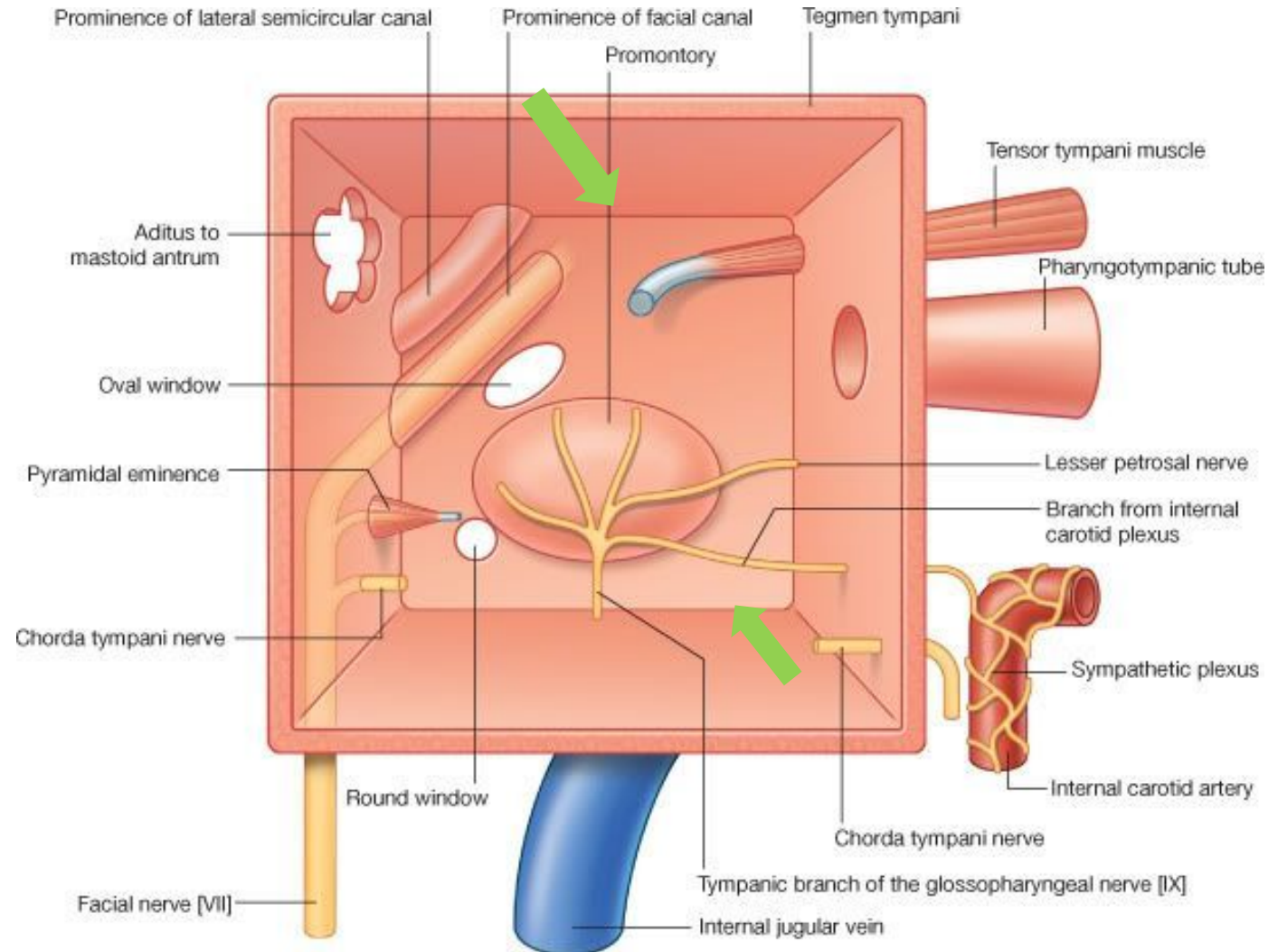
Boundaries of the middle ear:

1) Roof/Tegmental wall:

- It separates the middle ear from middle cranial fossa.
- It is formed by a thin plate of bone called tegmen tympani.

2) Floor/Jugular wall:

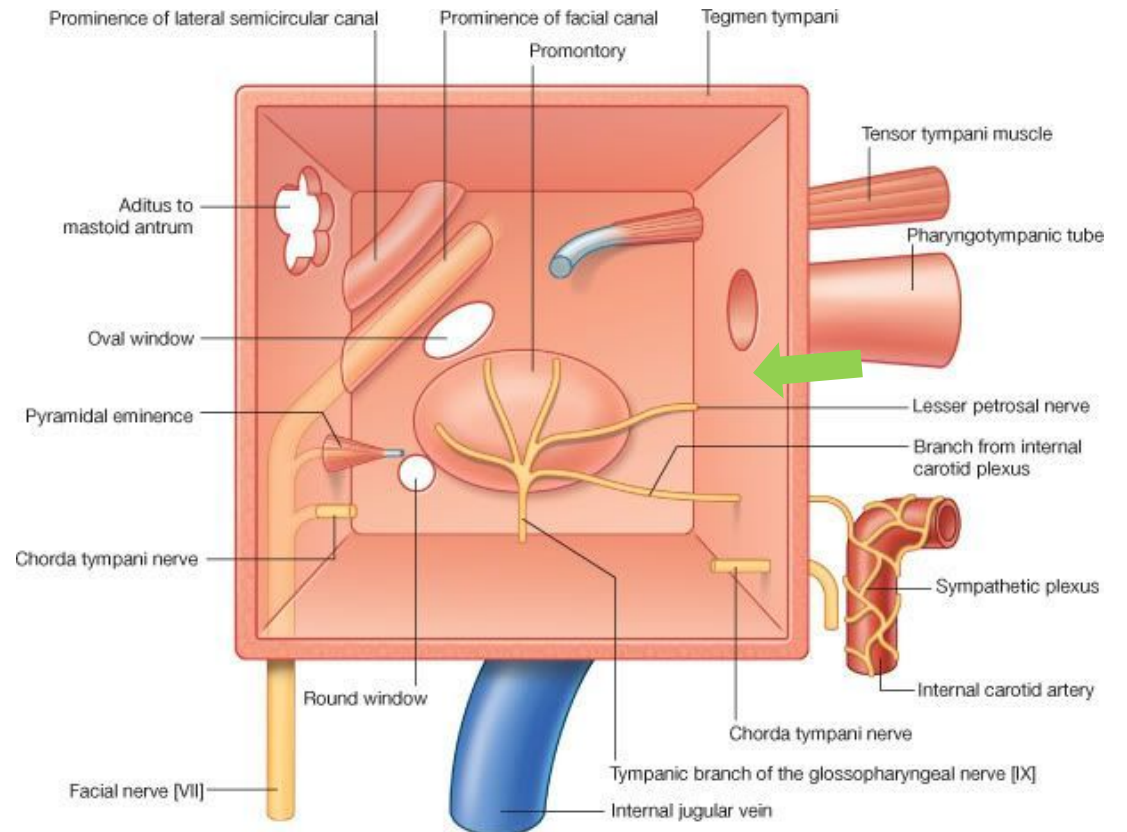
- It is formed by a thin plate of bone which separates middle ear from superior bulb of internal jugular vein.
- **Openings:** It has a canaliculus which transmits tympanic branch of glossopharyngeal nerve.



Boundaries of the middle ear:

3) Anterior/Carotid wall:

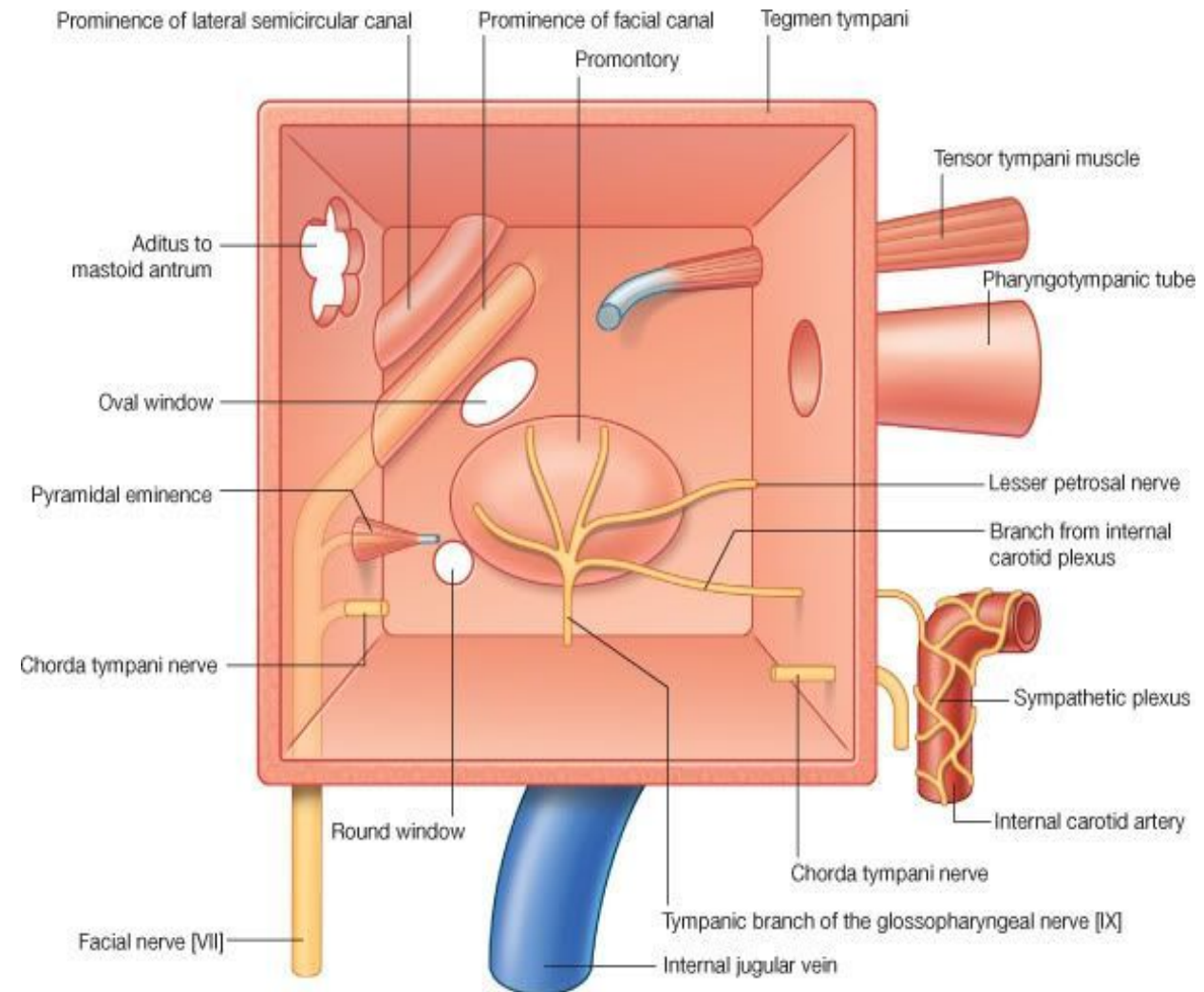
- It has the following features/openings from above downwards:
 - I. Opening for the canal of tensor tympani muscle
 - II. Opening of auditory tube
 - III. Perforations for the passage of superior and inferior sympathetic corticotympanic nerves.



Boundaries of the middle ear:

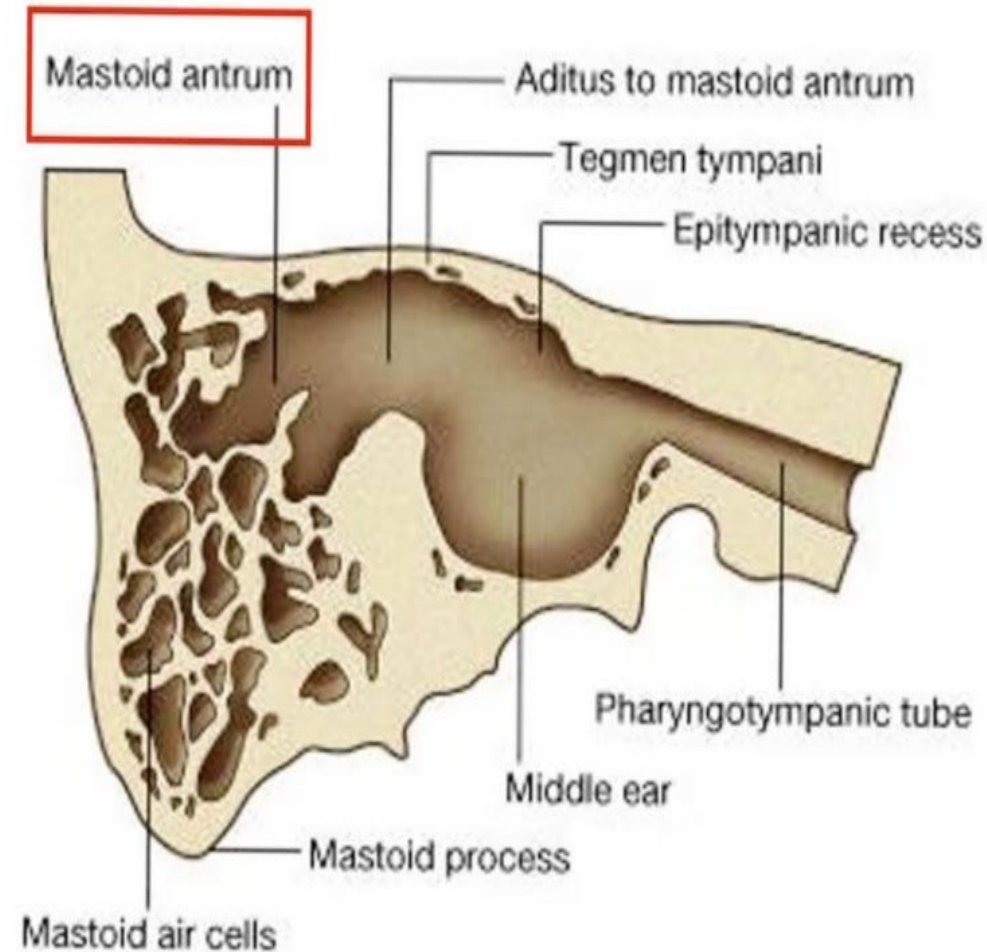
4) Posterior/Mastoid wall:

- It has the following features/openings from above downwards:
 - I. Aditus through which epitympanic recess communicates with mastoid antrum
 - II. Fossa incudis for short process of incus
 - III. A conical projection or pyramid with an opening for the passage of tendon of stapedius muscle
 - IV. A posterior canaliculus through which chorda tympani nerve enters middle ear cavity



Mastoid Antrum & Mastoid Air Cells

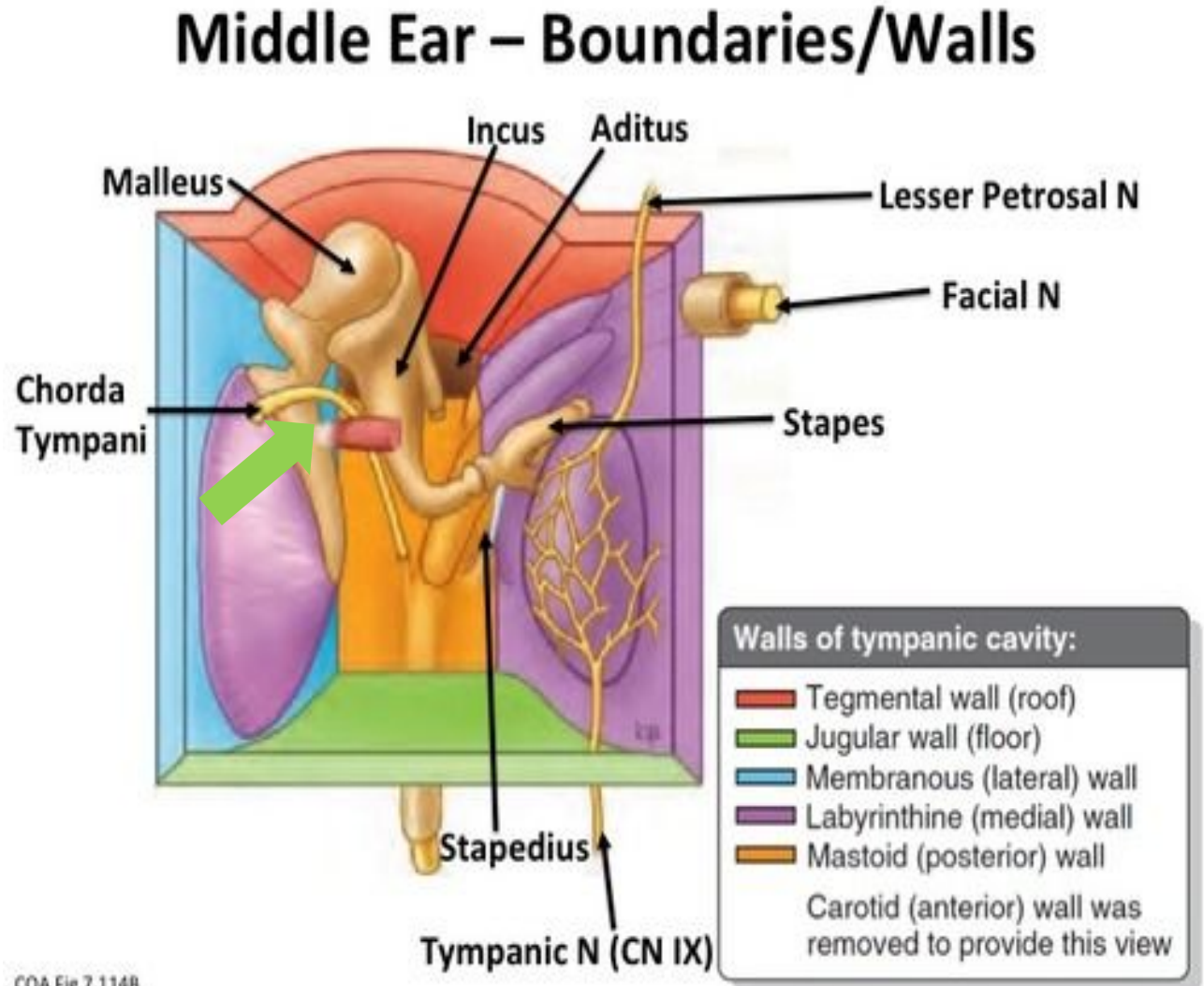
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Boundaries of the middle ear:

5) Lateral/membranous wall:

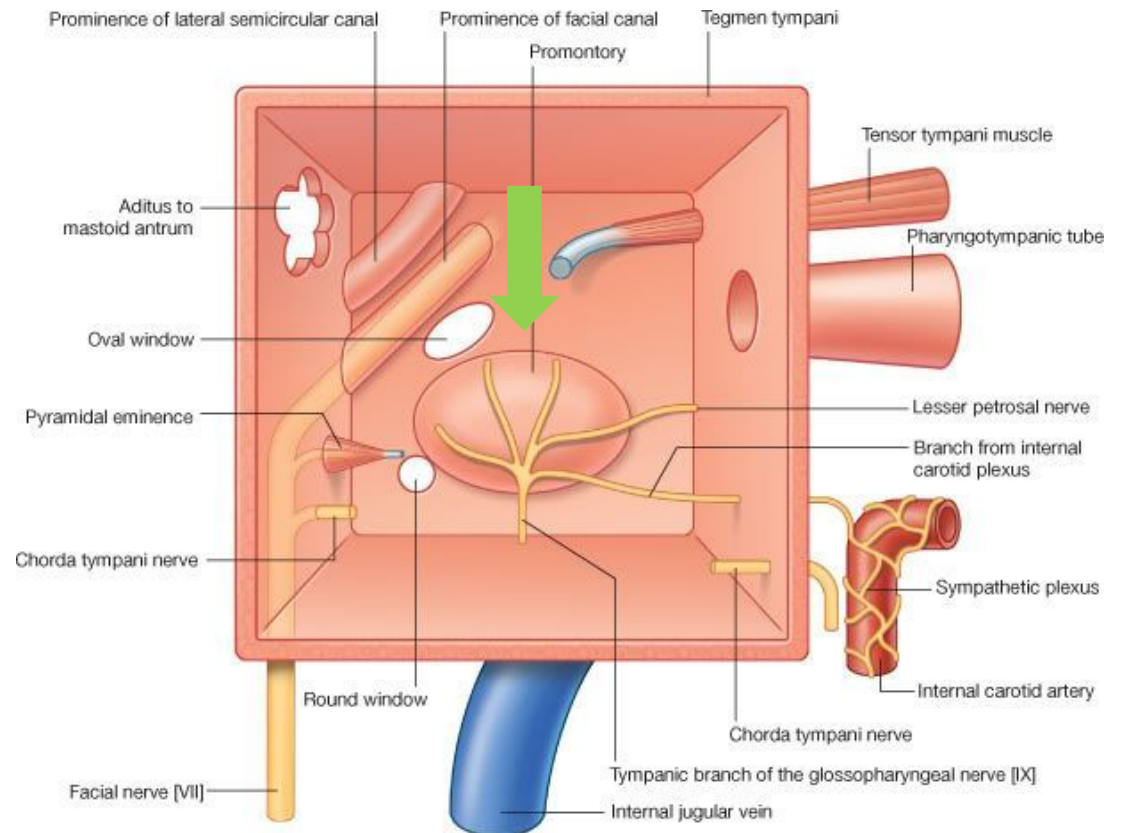
- It is mainly formed by tympanic membrane with tympanic sulcus and partly by squamous temporal bone.
- **Opening:** near the tympanic notch there is an anterior canaliculus for chorda tympani nerve to exit the middle ear.



Boundaries of the middle ear:

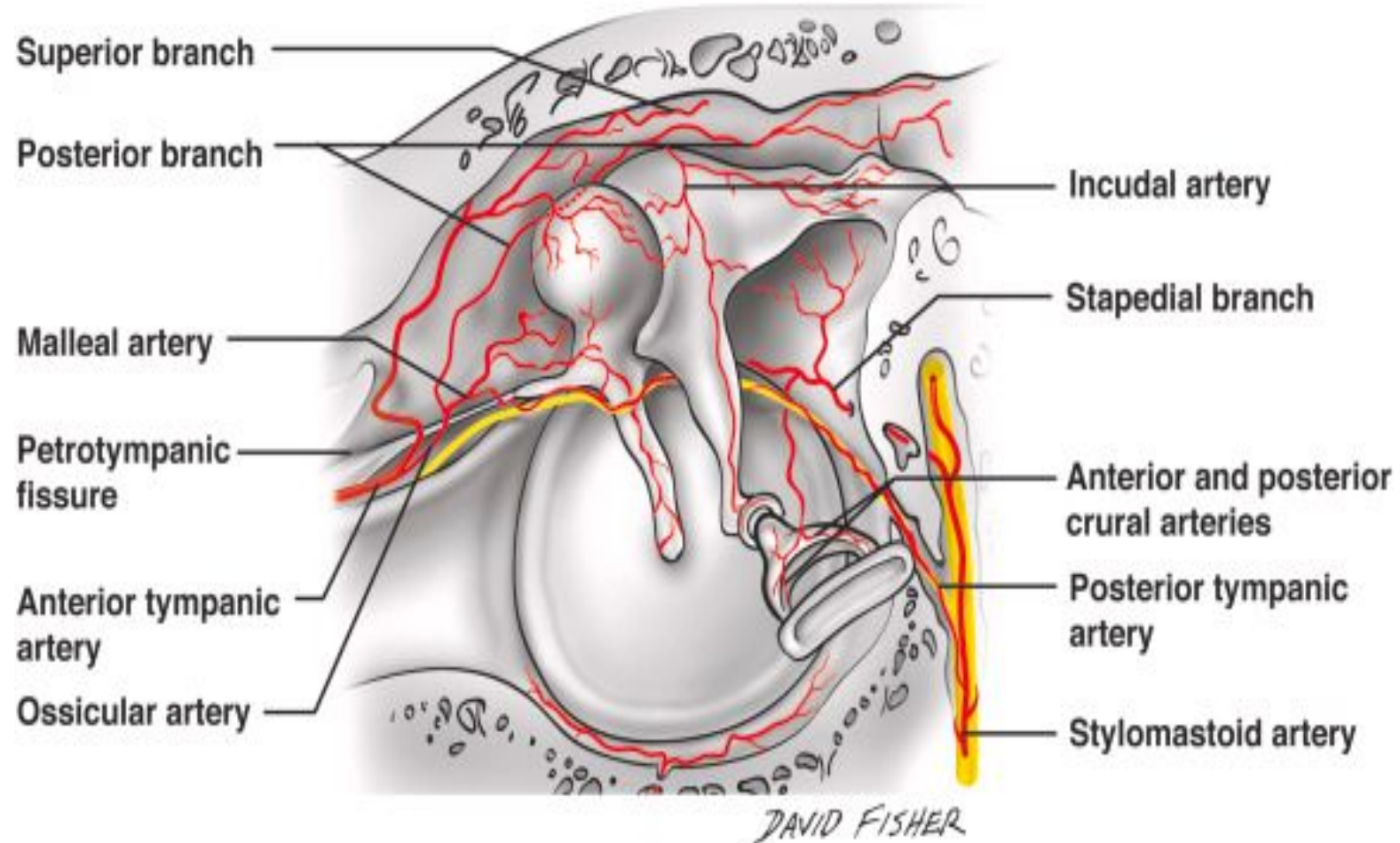
6) Medial/Labyrinthine wall:

- It separates the middle ear from the internal ear.
- **Features/openings:**
 - I. Promontory, a round bulging produced by cochlea
 - II. Oval window/fenestra vestibuli, lies posterosuperior to promontory which leads into vestibule of internal ear and closed by foot-plate of stapes
 - III. Facial canal, just above oval window
 - IV. Round window/fenestra cochleae, lies posteroinferior to promontory, opens into the cochlea and closed by secondary tympanic membrane



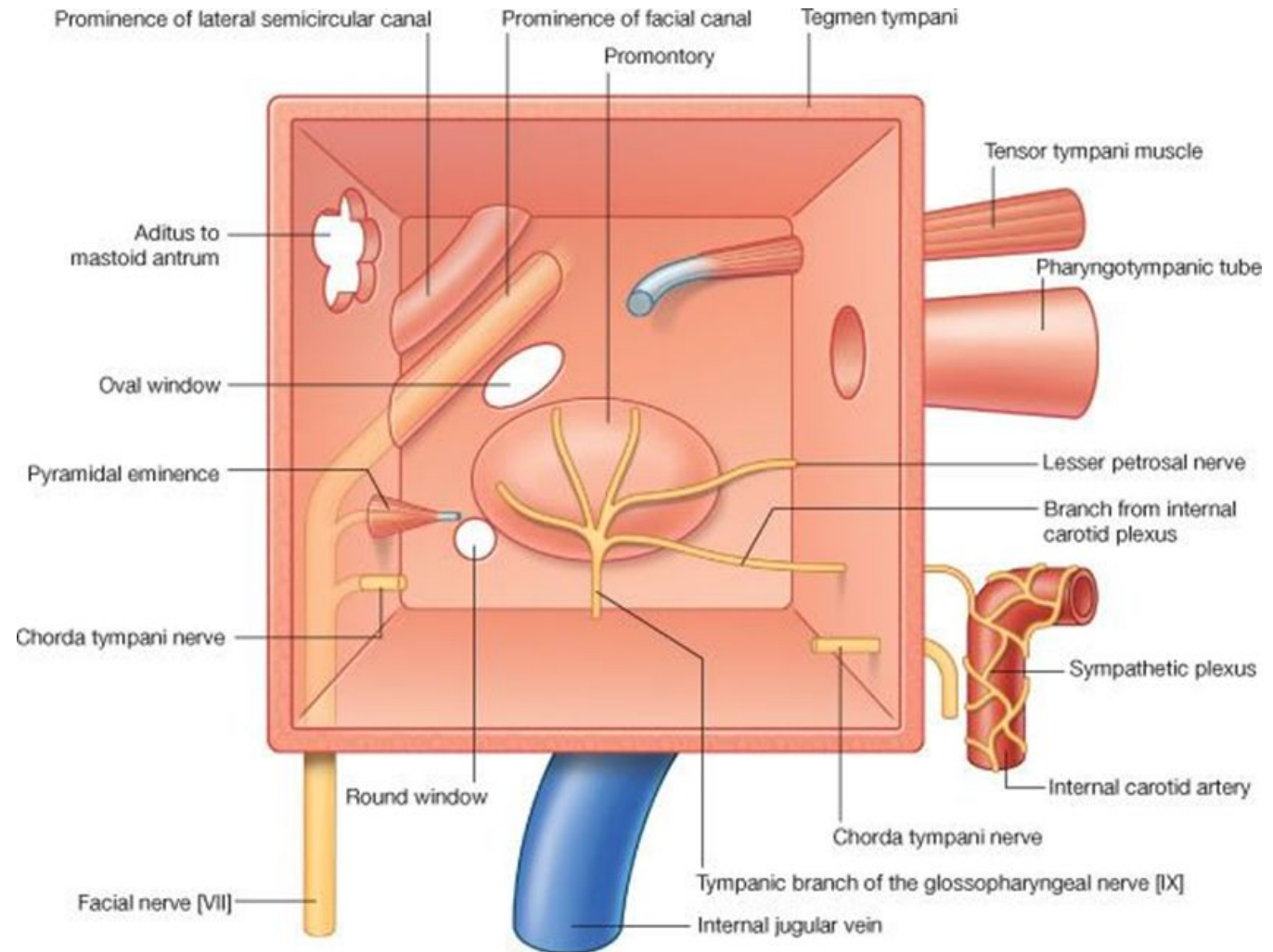
Blood supply & venous drainage of middle ear:

- Main arterial supply of middle ear is anterior tympanic branch of maxillary artery and posterior auricular artery.
- Veins drain into superior petrosal sinus and pterygoid venous plexus.



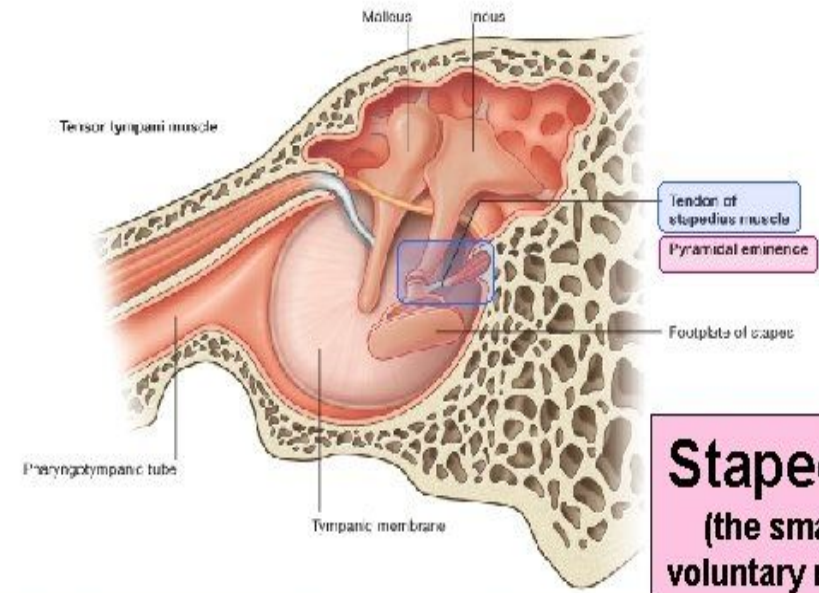
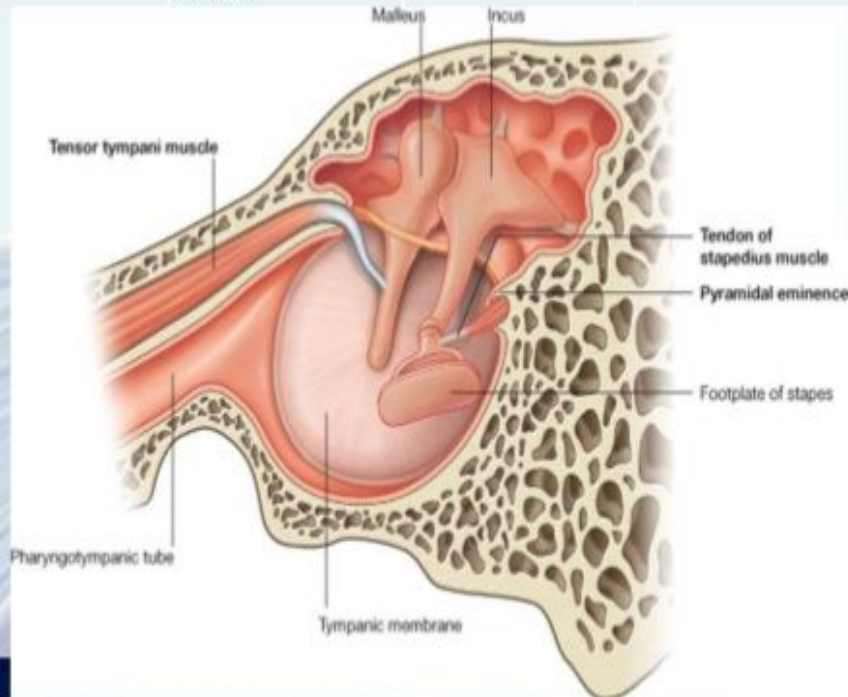
Nerve supply of the middle ear:

- Mainly from tympanic plexus, which lies over the promontory and is formed by:
 - I. Tympanic branch of glossopharyngeal nerve which supplies mucous membrane of middle ear, auditory tube, mastoid antrum and air cells.
 - II. Superior and inferior corticotympanic nerves arising from sympathetic plexus around internal carotid artery.



Muscles of the middle ear

MUSCLE	ORIGIN	INSERTION	NERVE SUPPLY	ACTION
Tensor tympani	Cartilaginous part of ET, its own bony canal	Upper part of handle of malleus	Branch from mandibular nerve [V ₃]	tensing tympanic membrane to reduce the force of vibrations in response to loud noises

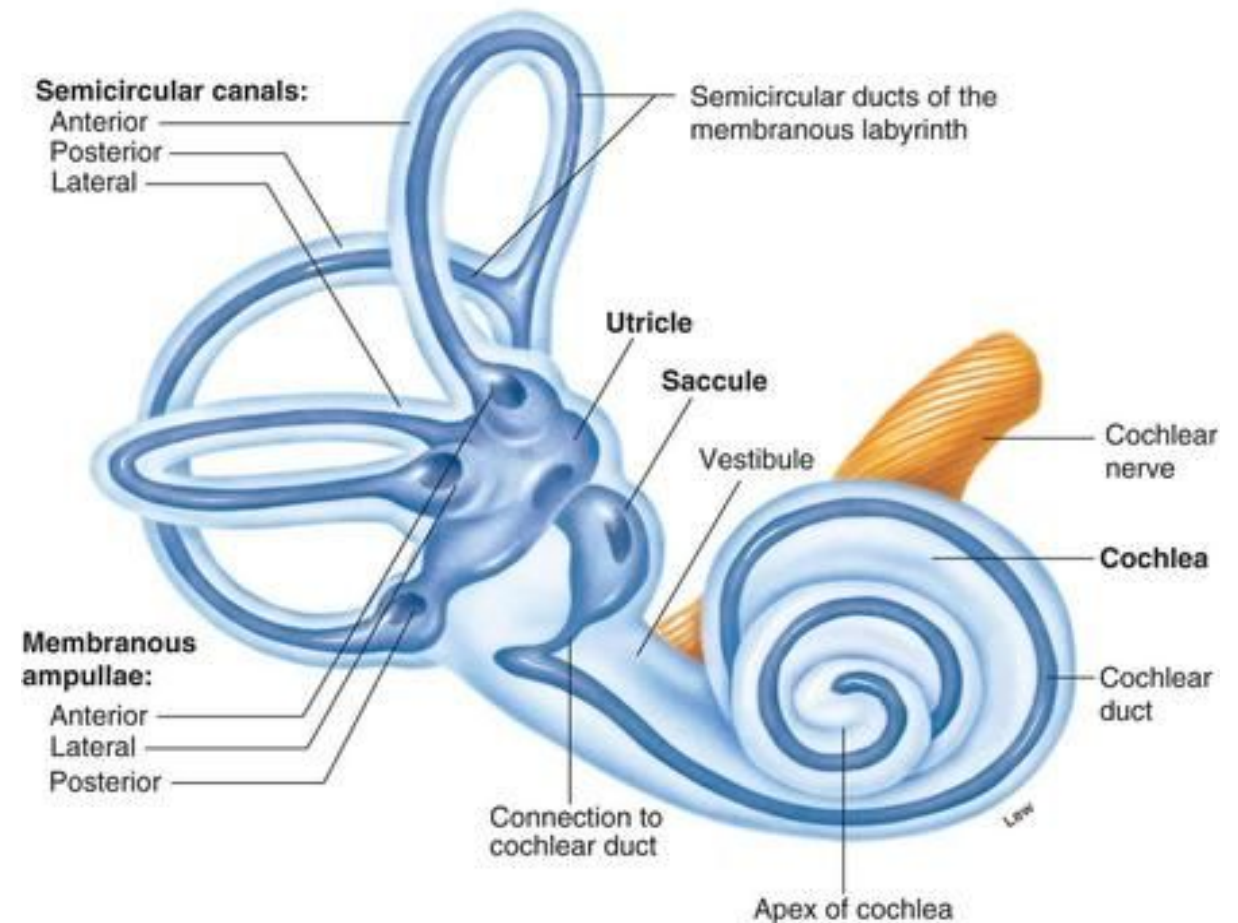


Stapedius
(the smallest voluntary muscle)

- **Origin:** Internal walls of the hollow pyramid.
- **Insertion:** The tendon emerges from the apex of the pyramid and is *inserted into the neck of the stapes*.
- **Nerve supply:** Facial nerve.
- **Action:** Reflexly damps down the vibrations of the stapes by pulling on the neck of that bone.

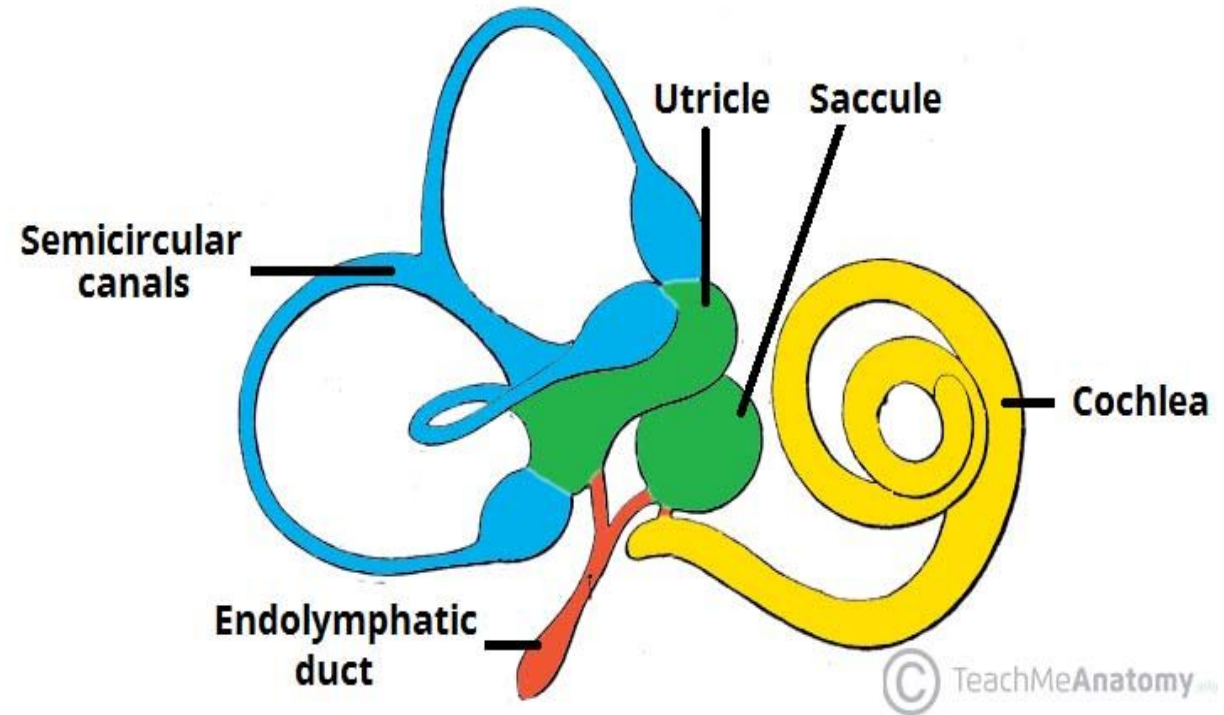
The Internal Ear (labyrinth):

- It lies within petrous temporal bone.
- Consists of a **bony labyrinth** and a **membranous labyrinth**.
- Membranous labyrinth contains a fluid called endolymph.
- Membranous labyrinth is separated from bony labyrinth by another fluid called perilymph.



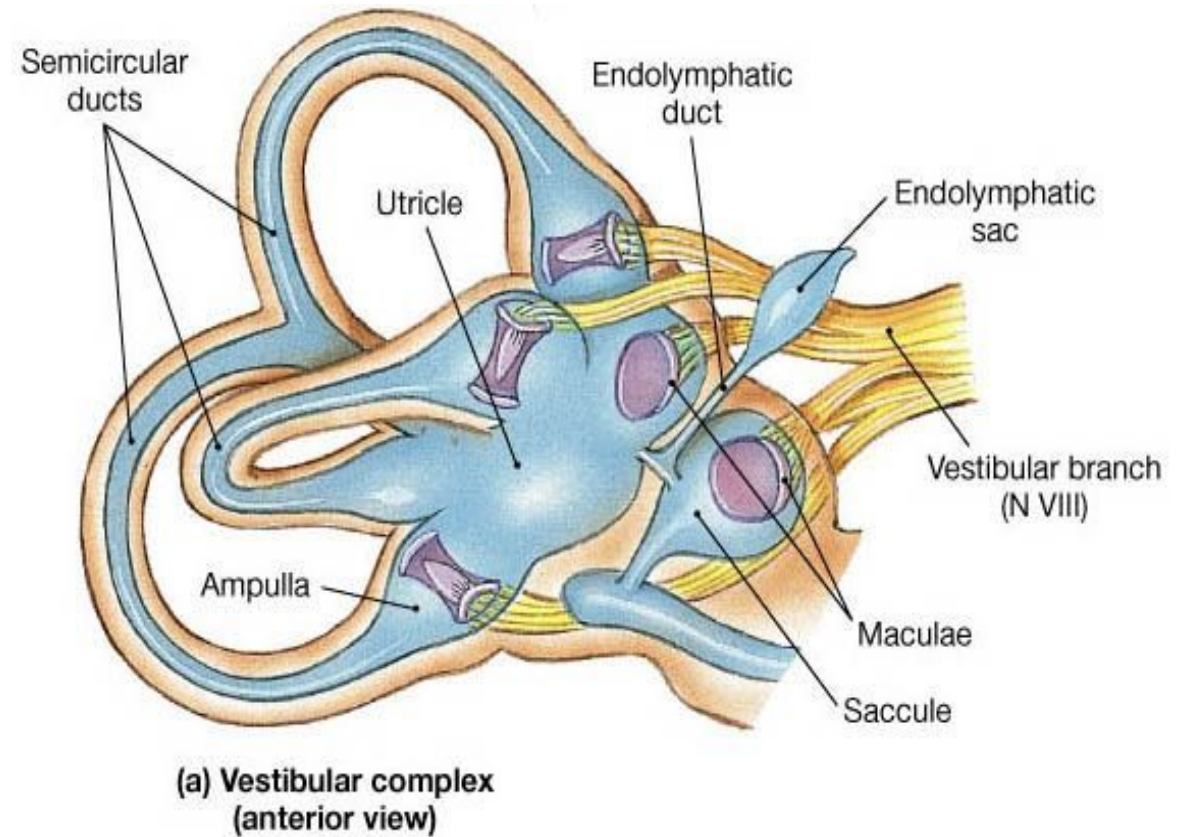
Bony Labyrinth:

- Consists of three parts viz, the cochlea anteriorly, vestibule in the middle and semi-circular canals posteriorly.
- I. **Cochlea:** resembles the shell of a snail, forms anterior part of labyrinth and the cochlear canal makes two and three quarter turns around a central axis.
 - II. **Vestibule:** it is the central part of bony labyrinth which opens into the middle ear at fenestra vestibuli. Three semicircular canals opens into it.
 - III. **Semicircular canals:** lie posteriorly, three in number, anterior, posterior and lateral. These three canals open into the vestibule by five openings.

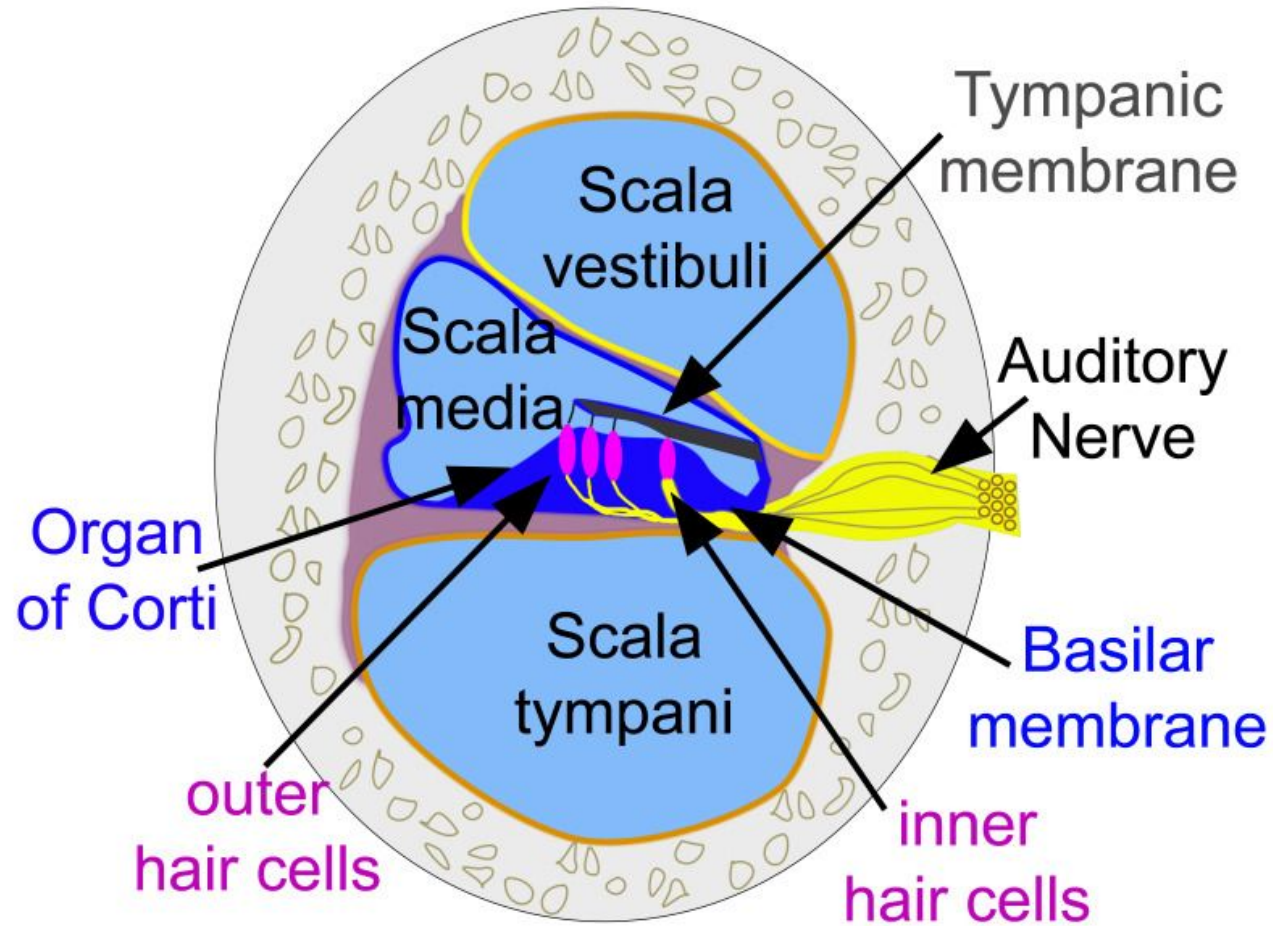


Membranous Labyrinth:

- It is a continuous closed cavity filled with endolymph.
- It contains **receptors for sound, the organ of Corti**, **for static balance, the maculae** & **for kinetic balance, the cristae**.
- It consists of three parts:
 - I. **Spiral duct of cochlea:** lies in cochlear canal and **contains spiral organ of Corti** which is end organ for hearing.
 - II. **Sacculle and utricle:** located in the vestibule, the sacculle is smaller and connected to cochlear duct whereas utricle is larger and connected to semicircular ducts. They **contain maculae, the static balance receptors**.
 - III. **Semicircular ducts:** lie in corresponding bony canals and **contain cristae, the kinetic balance receptors**.

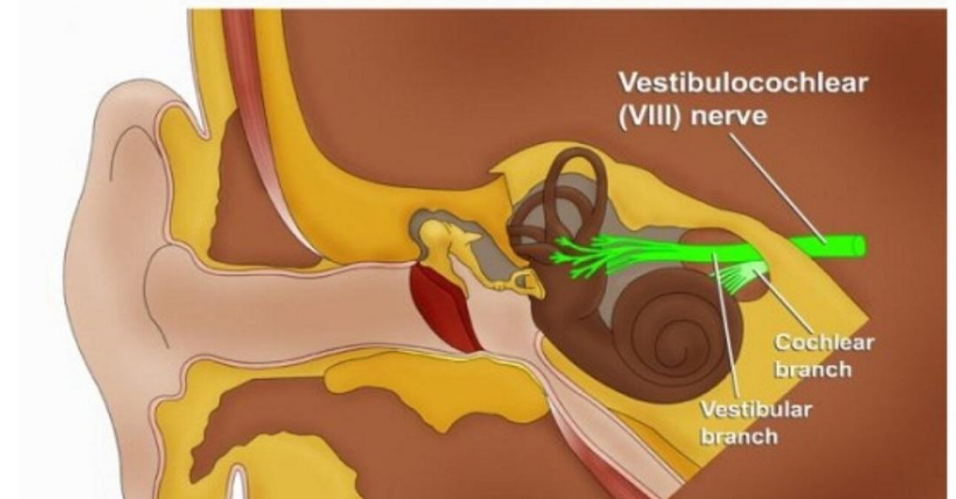
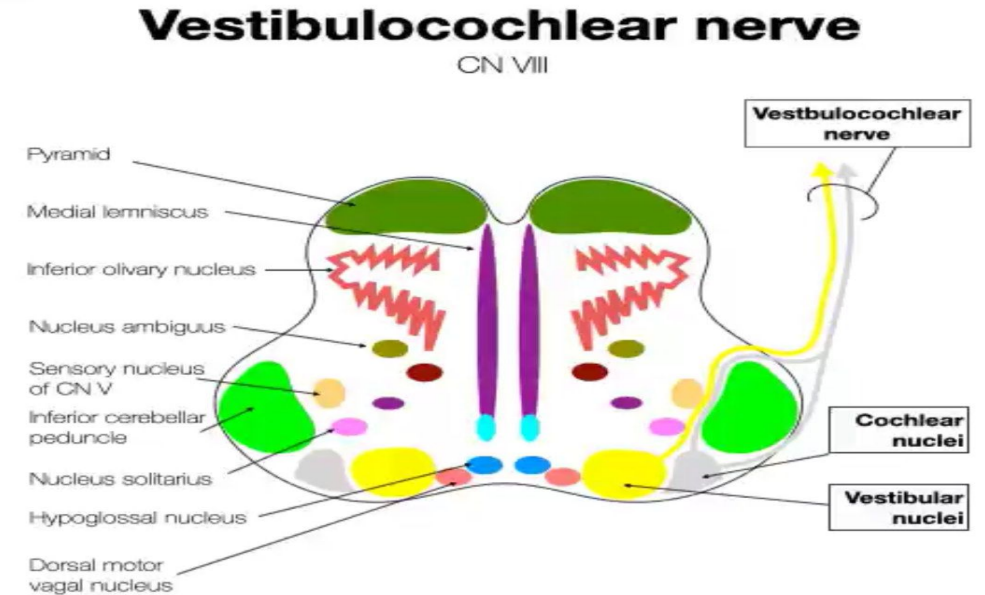


Section of cochlear canal illustrating its internal structure including the organ of Corti

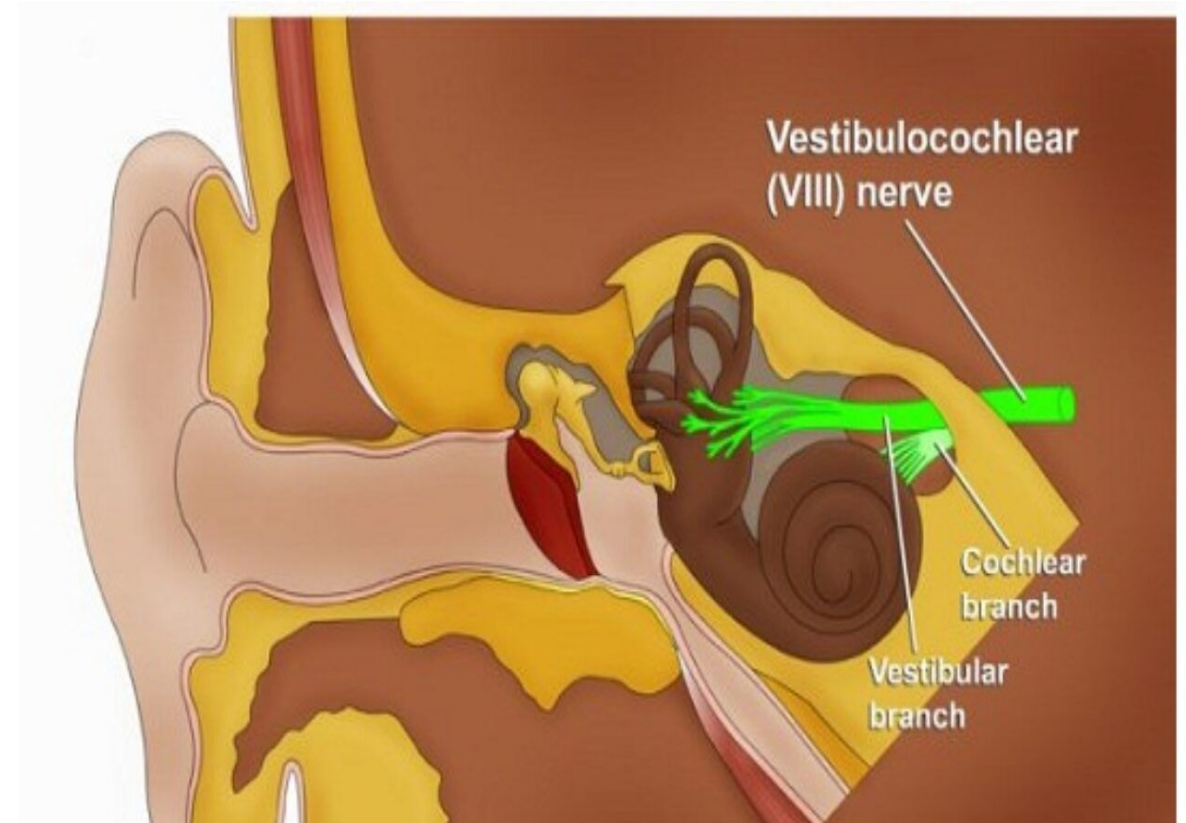


Vestibulocochlear nerve:

- It is the eighth cranial nerve and it is sensory in nature.
- It has two parts, the cochlear part concerned with hearing and a vestibular part concerned with equilibrium.
- **Nuclei:**
 - I. Dorsal and ventral cochlear nuclei related to inferior cerebellar peduncle
 - II. Superior, inferior, medial and lateral vestibular nuclei situated partly in medulla and partly in pons
- **Cochlear nerve:** conducts impulses from auditory receptors, the organ of Corti, situated in the cochlear duct.
- The auditory receptors are innervated by peripheral processes of bipolar neurons of spiral ganglion and the ganglion is situated in the cochlear canal.
- The central processes of these neurons form cochlear nerve which ends up in dorsal and ventral cochlear nuclei.



- **Vestibular nerve:** conducts impulses from vestibular receptors, the maculae of saccule & utricle for static balance and the cristae of semicircular ducts for kinetic balance.
- The vestibular receptors are innervated by peripheral processes of bipolar neurons of vestibular ganglion.
- The vestibular ganglion is situated in the internal acoustic meatus.
- The central processes of the bipolar neurons form vestibular nerve which ends up in vestibular nuclei.



Neurology of Hearing & Balance:

